

A nilpotent pure-part result for the Fong–Tsui problem

Run `fa_banach_001`, target arXiv:1511.09055

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Status. Partial result, likely valid. The result answers the finite nilpotent obstruction which appears in Mbekhta–Suciu’s analysis of the Fong–Tsui conjecture for contractive m -quasi-isometries. It does not settle the general Fong–Tsui conjecture for arbitrary pure contractions.

Source question

The source is Mostafa Mbekhta and Laurian Suciu, *Partial isometries and the conjecture of C. K. Fong and S. K. Tsui*, arXiv:1511.09055 [1]. They study bounded operators T on a complex Hilbert space satisfying

$$|T| \leq |\operatorname{Re} T|, \quad |T| = (T^*T)^{1/2}, \quad \operatorname{Re} T = \frac{T + T^*}{2}. \quad (1)$$

After proving the conjecture for contractive 2-quasi-isometries, they ask on page 11 whether the hypothesis $Q^2 = 0$ in Proposition 3.3 can be replaced by $Q^m = 0$ for some $m \geq 3$, in order to preserve the conclusion $Q = 0$ under condition (1).

Corollary 3.4. *A contractive 2-quasi-isometry which satisfies the condition (1.1) is self-adjoint.*

It is still unknown if the second assumption in the hypothesis of Proposition 3.3 namely $Q^2 = 0$, can be replaced by the weaker condition $Q^m = 0$ for some $m \geq 3$, in order to preserve the conclusion; that is to prove that $Q = 0$ under the condition (1.1).

However, for m -quasi-isometries we have the following

Corollary 3.5. *If $T \in \mathcal{B}(\mathcal{H})$ is a contractive m -quasi-isometry for an integer $m \geq 3$ which satisfies the condition (1.1) then $\mathcal{N}(I - T^*T) = \mathcal{N}(I - TT^*) = \overline{\mathcal{R}(T^m)}$ reduces T to a symmetry. In addition, T is self-adjoint if and only if $T = S \oplus 0$.*

Proof. If T is a m -quasi-isometry then $T|_{\mathcal{R}(T^m)}$ is an isometry. Such an operator has the form (3.7) on $\mathcal{H} = \overline{\mathcal{R}(T^m)} \oplus \mathcal{N}(T^{*m})$ with S an isometry and $Q^m = 0$. In the case when $\|T\| = 1$ we also have (as in the proof of Proposition 3.3) that $S^*R = 0$ and

$$\mathcal{N}(I - T^*T) = \mathcal{N}(I - TT^*) = \overline{\mathcal{R}(T^m)} \oplus \mathcal{N}(I - QQ^*),$$

Proof intuition

The finite nilpotent case has a numerical-radius obstruction which is not used in the source proof. If $N^m = 0$, then the Haagerup–de la Harpe theorem gives a strict upper bound

$$w(N) \leq \|N\| \cos \frac{\pi}{m+1} < \|N\|$$

for nonzero N , after normalizing. On the other hand, the Fong–Tsui inequality $|N| \leq |\operatorname{Re} N|$ forces

$$\|N\| = \||N|\| \leq \||\operatorname{Re} N|\| = \|\operatorname{Re} N\| \leq w(N).$$

Thus a nonzero nilpotent would have numerical radius simultaneously at least and strictly less than its norm. The only possibility is $N = 0$.

Main result

Theorem 1 (Finite nilpotent blocks vanish). *Let $N \in \mathcal{B}(\mathcal{H})$ be a bounded operator on a complex Hilbert space. If $N^m = 0$ for some integer $m \geq 2$ and*

$$|N| \leq |\operatorname{Re} N|,$$

then $N = 0$.

Proof. Assume $N \neq 0$. Since $0 \leq |N| \leq |\operatorname{Re} N|$, monotonicity of the norm on positive operators gives

$$\|N\| = \||N|\| \leq \||\operatorname{Re} N|\| = \|\operatorname{Re} N\|.$$

For a self-adjoint operator A , $\|A\| = \sup_{\|x\|=1} |\langle Ax, x \rangle|$. Applying this to $A = \operatorname{Re} N$,

$$\|\operatorname{Re} N\| = \sup_{\|x\|=1} |\operatorname{Re} \langle Nx, x \rangle| \leq \sup_{\|x\|=1} |\langle Nx, x \rangle| = w(N),$$

where $w(N)$ is the numerical radius.

The Haagerup–de la Harpe numerical-radius inequality for nilpotents [2] says that if $N^m = 0$, $m \geq 2$, then

$$w(N) \leq \|N\| \cos \frac{\pi}{m+1}.$$

Combining the inequalities yields

$$\|N\| \leq w(N) \leq \|N\| \cos \frac{\pi}{m+1}.$$

Since $0 < \cos(\pi/(m+1)) < 1$, this contradicts $\|N\| > 0$. Hence $N = 0$. $\square$

Corollary 1 (Replacement in Proposition 3.3). *In Proposition 3.3 of [1], the alternative hypothesis $Q^2 = 0$ may be replaced by $Q^m = 0$ for any finite $m \geq 2$. More precisely, let*

$$T = \begin{pmatrix} S & R \\ 0 & Q \end{pmatrix} \quad \text{on } \mathcal{H} = \mathcal{G} \oplus \mathcal{G}^\perp,$$

where $\|T\|^{-1}S$ is an isometry on $\mathcal{G}$, and assume $Q^m = 0$ for some finite $m \geq 2$. If T satisfies (1), then T is self-adjoint.

Proof. If $T = 0$, there is nothing to prove. Otherwise scale as in the proof of [1, Proposition 3.3], writing $T_0 = \|T\|^{-1}T$, $S_0 = \|T\|^{-1}S$, $R_0 = \|T\|^{-1}R$, and $Q_0 = \|T\|^{-1}Q$. The initial part of that proof uses Theorem 3.1 of the source paper and does not use $Q^2 = 0$. It gives

$$R_0 R_0^* = P_{\mathcal{N}(S_0^*)}, \quad Q_0 R_0^* = 0,$$

and, in the nilpotent alternative, that $\mathcal{N}(I - Q_0 Q_0^*)$ is invariant under Q_0^* .

If $b \in \mathcal{N}(I - Q_0 Q_0^*)$, then $b = Q_0 Q_0^* b$. Since this nullspace is Q_0^* -invariant, an induction gives

$$b = Q_0^k Q_0^{*k} b \quad (1 \leq k \leq m).$$

Taking $k = m$ and using $Q_0^m = 0$, we obtain $b = 0$. Hence $\mathcal{N}(I - Q_0 Q_0^*) = \{0\}$. The remainder of the reduction in the source proof then yields $R_0 = 0$, so $T_0 = S_0 \oplus Q_0$, and both summands satisfy (1).

Theorem 1 applied to Q_0 gives $Q_0 = 0$. The same source argument gives that S_0 is a symmetry. Thus $T = \|T\|(S_0 \oplus 0)$ is self-adjoint. $\square$

Remark 1 (Blockwise inheritance). *The last step uses the elementary fact that if $T = S \oplus Q$ is a reducing direct sum and T satisfies (1), then Q also satisfies (1). Indeed,*

$$|S \oplus Q| = |S| \oplus |Q|, \quad |\operatorname{Re}(S \oplus Q)| = |\operatorname{Re} S| \oplus |\operatorname{Re} Q|.$$

Restricting the positive operator inequality to the Q -summand gives $|Q| \leq |\operatorname{Re} Q|$.

Corollary 2 (Contractive finite m -quasi-isometries). *Let $T \in \mathcal{B}(\mathcal{H})$ be a contractive m -quasi-isometry for a finite integer $m \geq 1$. If T satisfies (1), then T is self-adjoint.*

Proof. For $m = 1$ and $m = 2$, this is already contained in the source paper: see the quasi-isometry case and Corollary 3.4 of [1]. Let $m \geq 3$. Corollary 3.5 of [1] shows that

$$\mathcal{H} = \overline{\mathcal{R}(T^m)} \oplus \mathcal{N}(T^{*m})$$

reduces T to the block diagonal form

$$T = S \oplus Q,$$

where S is a symmetry on $\overline{\mathcal{R}(T^m)}$ and $Q^m = 0$. Since (1) holds for T , the preceding blockwise inheritance gives $|Q| \leq |\operatorname{Re} Q|$. Theorem 1 gives $Q = 0$. Hence $T = S \oplus 0$, so $T = T^*$. $\square$

Verification notes and limitations

The proof is analytic and uses no finite search. The two main points for review are:

1. the Haagerup–de la Harpe bound applies to arbitrary bounded Hilbert-space nilpotents after scaling by $\|N\|$;
2. in a reducing direct sum $T = S \oplus Q$, the operator inequality $|T| \leq |\operatorname{Re} T|$ restricts to each summand.

This packet does not prove the full Fong–Tsui conjecture and does not address general pure contractions whose pure part is not finite nilpotent. Corollary 1 closes the finite nilpotent obstruction left open on page 11 of the source paper and upgrades the source’s contractive m -quasi-isometry conclusion from “ T is self-adjoint if and only if $T = S \oplus 0$ ” to self-adjointness for every finite m .

Novelty check

Local run indexes were searched for arXiv:1511.09055, the title terms, “Fong–Tsui”, $Q^m = 0$, nilpotent pure part, and Haagerup numerical radius; no existing packet or attempt for this target was found. Web searches for “Fong-Tsui conjecture nilpotent”, “Fong-Tsui $Q^m = 0$ ”, “ $|T| \leq |\operatorname{Re} T|$ nilpotent”, and the Haagerup–de la Harpe nilpotent numerical-radius theorem found the source paper, Mortad’s earlier contribution, and numerical-radius references, but no later explicit answer to the page-11 nilpotent question.

References

- [1] M. Mbekhta and L. Suci, *Partial isometries and the conjecture of C. K. Fong and S. K. Tsui*, arXiv:1511.09055.
- [2] U. Haagerup and P. de la Harpe, *The numerical radius of a nilpotent operator on a Hilbert space*, Proceedings of the American Mathematical Society **115** (1992), 371–379.
- [3] M. H. Mortad, *A contribution to the Fong–Tsui conjecture related to self-adjoint operators*, arXiv:1208.4346.